

**Calibre DRC of UMC 0.13um Logic and
Mixed-Mode 1P8M MMC AL Enhancement
Shrinkable Process
Application Note**

Revision History

Revision	Date	Editor	Description
2.5_P1	06/16/2020	D.C Wu	1.Add: 4C.1J? 2.Update: umc_ant_LOGIC_MIXED-MODE13_1PxMyT-MM C-AL-L130E-SHRINKABLE-CALIBRE-DRC-0.1_P 2.cal 3.Delete: 4C.1J, 4C.1K
2.3_P1	03/18/2020	D.C Wu	1.Add: umc_ant_LOGIC_MIXED-MODE13_1PxMyT-MM C-AL-L130E-SHRINKABLE-CALIBRE-DRC-0.1_P 1.cal 2.Delete: umc_ant_LOGIC_MIXED-MODE13_1PxMyT-MM C-AL-L130E-SHRINKABLE-CALIBRE-DRC-v8.cal
2.2_P1	02/03/2020	D.C Wu	1.Add: 4C.1J, 4C.1K 2.Update: 4C.1F, 110nm_layers_v1.8_p5.cal
2.1_P1	09/26/2019	D.C Wu	1.Update version only
2.0_P1	07/23/2018	Todd Hung	1.Update: 4B.2E, 4B.4E_Vln(n=1X small via), 4B.5E_Vln(n=2X large via)
1.10_P2	04/25/2018	Todd Hung	1.Update 4C.8.1.Bb
1.10_P1	06/09/2017	Todd Hung	1.Add 4A.Note3
1.9_P1	01/20/2017	Todd Hung	1.Update 5.3T, 5.3W
1.8_P1	4/14/2016	Janice Tseng	1.Update 4B.5D_VI?, 4C.1I
1.7_P1	12/21/2015	C.C. Hung	1.Update 4A.1I, 4B.2D, 4B.4D 2.Add 4A.10E_11E.Note
1.6_P1	07/06/2015	Todd Hung	1.Update 4B.5D_VI? 2.Add 4C.1I
1.5_P2	03/02/2015	C.T. Lai	1.Update 4C.8.2D.b
1.5_P1	12/18/2014	C.T.Lai	1.Update 4B.4B.c_VI?, 4B.5B.c_VI? 2.Add 4C.1H
1.3_P1	10/02/2014	C.T. Lai	1.Update 4B.2B.c.VI1, 4C.8.1K, 4C.8.1L 4C.8.2Ja, 4C.8.2K?, 4C.8.2L?, 4C.8.2P? 2.Add 4C.8.2R 3.Delete 4C.8_Note2
1.2_P1	08/06/2014	C.T. Lai	1.Add 4A.9D.a3_5V, 4A.9D.b3_5V, 4A.9D.c3.1_5V, 4A.9D.c3.2_5V 2.Modify 4A.13C.b_5V, 4C.8.1.Ab, 4C.8.1.Bb 4C.8.1Nb, 4C.8.1Nc, 4C.8.2Pb, 4C.8.2Pc
1.1_P1	03/12/2014	C.T. Lai	1.Delete 4A.15L, 4A.15M 2.Change rule ID : 4A.15F~K to 4A.15E~J 3.Update SRAM version
1.0_P2	10/22/2013	C.T. Lai	1.Update definition of P_sub for 4A.15J
1.0_P1	8/22/2013	C.T. Lai	1.Update layer mapping 2.MMC should be place between thin and thick metal 3.Update 4A.1E, 4A.1G?, 4A.11M, 4C.2D, 4C.3E 4C.2F, 4C.2G, 4C.2H, 4C.4C, 4C.4E, 4A.15J 4.Delete 4A.15E,5.3X.a 5.Add 4C.8? 6.Change rule ID: 5.3X.b to 5.3X.a

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			7.Update metal slot & antenna rule
0.4_P3	2/21/2013	C.T. Lai	1.Set Switch of Grid size (Doc No.LCJC-94N8UP)
0.4_P2	1/2/2013	C.T. Lai	1. Update for the SRAM CELL (Doc No.ATSI-93DEXB
0.4_P1	8/6/2012	Eric Kuo	1. Add switch 4A.Note2_Recommended 2. Modify switch Recommend_PO1_CHECK 3. Add switch CHECK_5V_IO
0.3_P1	1/4/2012	K.C. Hsu	1. Add switch CHECK_3.3V_IO_COMMON 2. Modify switch CHECK_RECOMMEND
0.2_P1	8/23/2011	K.C. Hsu	1. Rename switch from DIFFUSION_DENSITY to DIFF_DENSITY; from CHECK_REGULAR_IO to CHECK_3.3V_REGULAR_IO; from CHECK_HG_IO to CHECK_3.3V_HG_IO 2. Add switch CHECK_RECOMMEND
0.1_P3_03	7/26/2010	K.C. Hsu	Update Die Seal Ring Rule
0.1_P3_02	6/7/2010	K.C. Hsu	Original Release

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Chapter 1. Switch and Option Usage

To match different usages from users, this DRC run deck is coded with several kinds of switch and option. User can find the switches started by `//DRC UNSELECT CHECK xxxxxx` or `//#DEFINE xxxxxx`.

The switch started by `//DRC UNSELECT CHECK xxxxxx` is to turn ON/OFF some rules, the marker `//` implies this line is commented out and the following rule xxxxxx will be checked. We call this switch with `//` for `DRC UNSELECT CHECK` is ON. On the other hand, the switch `DRC UNSELECT CHECK` without `//` is called by OFF and rule xxxxxx will not be checked.

The option started by `//#DEFINE xxxxxx` is to define some options, the marker `//` implies this line is commented out and the following option xxxxxx will be selected. We call this option without `//` for `#DEFINE` is ON. On the other hand, the option `#DEFINE` with `//` implies this line is commented out and is called by OFF. By this setting, the option xxxxxx will not be selected.

Please refer to 1-1.Switch and Options Information for details.

1-1. Switch and Option Information

(1) DRC UNSELECT CHECK

DRC UNSELECT CHECK DIFF_DENSITY

Switch for diffusion density rule. This switch is off as default

DRC UNSELECT CHECK POLY_DENSITY

Switch for poly density rule. This switch is off as default

DRC UNSELECT CHECK METAL_DENSITY

Switch for metal density rule. This switch is off as default

DRC UNSELECT CHECK SEAL_RING_RULES

Switch for die seal ring rule. This switch is off as default

DRC UNSELECT CHECK MAX_MET_SPACE

Switch for maximum spacing of metal rule. This switch is on as default

DRC UNSELECT CHECK Recommend_SAB_CHECK

Switch for Recommend SAB rule. This switch is off as default

DRC UNSELECT CHECK Extra_CHECK

Switch for remindful checking. This switch is off as default.

DRC UNSELECT CHECK ILLEGAL_DUMMY

Switch for simple dummy checking. This switch is off as default.

DRC UNSELECT CHECK Recommend_PO1_CHECK

Switch for voltage drop rules checking. This switch is on as default.

DRC UNSELECT CHECK CHECK_2.5V_IO

Switch for 2.5V I/O checking. This switch is on as default.

DRC UNSELECT CHECK CHECK_3.3V_IO

Switch for 3.3V I/O rule checking. This switch is off as default.

DRC UNSELECT CHECK CHECK_5V_IO

Switch for 5V I/O rule checking. This switch is off as default.

DRC UNSELECT CHECK CHECK_3.3V_REGULAR_IO

Switch for 3.3V REGULAR I/O checking. This switch is on as default.

DRC UNSELECT CHECK CHECK_3.3V_HG_IO

Switch for 3.3V HG I/O checking. This switch is off as default.

DRC UNSELECT CHECK OFFGRID_5

Switch for Grid Size 0.005 checking. This switch is off as default.

(2) #DEFINE OPTION

Please set up the appropriate MMC option which your chip implemented, and please DO NOT set two or more MMC options

```
=====
MMC customer's definition
=====
```

```
#DEFINE M67_MMC (for 1P7M process)
//#DEFINE M56_MMC
//#DEFINE M45_MMC
//#DEFINE M34_MMC
//#DEFINE M23_MMC
```

Chapter 2. Environment Setting

To ensure the DRC command file is executed correctly, please selecting the correct METAL-OPTION and setting the environment of run deck before using this command file. Please refer to the following sections as important notices.

2-1. Layout Environment Set Up

Please set up RESOLUTION, LAYOUT PATH and LAYOUT PRIMARY according to your design before using CALIBRE DRC.

2-2. Layer Mapping Set Up

Please fill in the layer number information in layer mapping file, Incorrect layer mapping information will cause serious mistake.

2-3. Metal option Set Up

This DRC package includes several run sets separated by different metal structure. Please refer to the following table to select the necessary run set.

Metal Option	ME8	VI7	ME7	VI6	ME6	VI5	ME5	VI4	ME4	VI3	ME3	VI2	ME2	VI1	ME1
1P4M									O	O	O	O	O	O	O
1P5M							O	O	O	O	O	O	O	O	O
1P6M					O	O	O	O	O	O	O	O	O	O	O
1P7M			O	O	O	O	O	O	O	O	O	O	O	O	O
1P8M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O

Chapter 3. Special Rule Check

3-1. Antenna rule check

Due to the special coding style, DRC for antenna rule check is separated by an independent file in this DRC command file package with the file name started by: umc_ant_xxxxxx. Please setup the correct environment and run it separately.

3-2. Logo rule check

Most of rule checks are excluded in logo area and additional logo rule check is applied in this run set.

3-3. Memory rule check

Since the SRAM cell is taken care manually, some logic rules are waived in SRAM areas. Logic rule check is excluded sometimes and some specified SRAM rules are implemented in this run set. It is better to contact your contact window in UMC to ensure the SRAM cell you got is applied in this run set before using it.

3-4. Global Rules

Global rules are **NOT** included in this rule package. Please refer to the DSM (Design Support Manual) and find out the corresponding TLR and DRC command file.